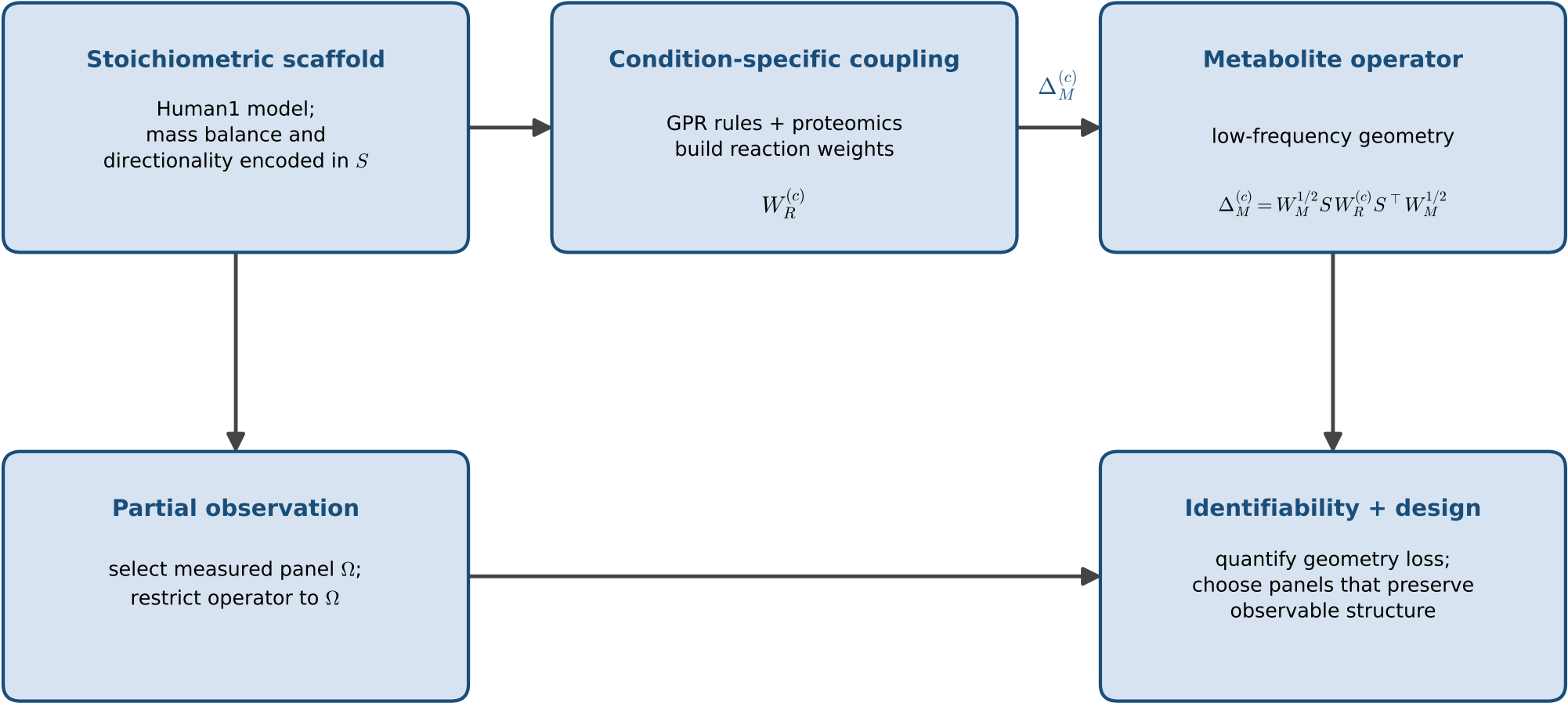


Figure 1: Operator-centric workflow for metabolite-focused geometry



The bipartite graph illustrates metabolite–reaction incidence; mass balance and reaction directionality are encoded by the stoichiometric matrix S , not by the graph itself.